Fiscal Year:	FY 2015	**Task Last Updated:**	FY 10/12/2015
PI Name:	Wotring, Virginia Ph.D.		
Project Title:	Are Medications Involved in Vision and Intracranial Pressure Changes Seen in Spaceflight?		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77030	**Congressional District:**	9
Comments:	PI formerly with Universities Space Research Association until fall 2015.		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	04/04/2012	**End Date:**	06/30/2015
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Norsk, Peter	**Contact Phone:**	
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Flight Program:	ISS		
Flight Assignment:	NOTE: original end date was 12/31/2012; extended to 6/30/2015 per HRP management (Ed., 7/9/14)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	It has recently been reported that intracranial pressure (ICP) and/or vision changes have occurred in a number of long duration astronauts. The reasons for these potentially debilitating medical issues are currently unknown. The potential role of administered medications has not yet been examined, but it is known that in terrestrial medical practice, some medications may raise ICP and/or cause visual changes. This project's purpose is to determine if affected crewmembers used medications known to be associated with visual changes and ICP elevations in terrestrial medical practice and to assess the likelihood of medications as causal agents in spaceflight-associated visual changes and ICP elevation by using a large terrestrial database to model correlations.		

Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research, which requires focused and constrained data gathering and analysis that is more appropriately obtained through a non-competitive proposal.
Research Impact/Earth Benefits:	
Task Progress:	In this final project year, Lifetime Surveillance of Astronaut Health (LSAH) completed consenting crewmember subjects for this study, and delivered the final data set. Only 2 subjects who were approached to participate in this state declined or were unresponsive. Data from the other 39 subjects were compiled and analyzed. Work was slower than expected due to the departure of data analyst Shahin Pour at the end of 2014. A suitable replacement was not found until July 2015. Prior to delivery of the final identifiable, consented data set, LSAH had provided an incomplete and de-identified data set. A publication was prepared from those data: Wotring VE. “Medication use by U.S. crewmembers on the International Space Station.” FASEB J. 2015 Jul 17. (See Bibliography section). This was the first publication regarding medication use on long duration missions. Descriptive data analysis has been completed, and results are described below. Briefly, we examined the medication use records of all consenting crewmembers who have had a CPG score assigned to them. The subjects included males and females, International Space Station (ISS) crew, and shuttle crew, Vision Impairment and Intracranial Pressure (VIIP) cases and confirmed non-cases. As expected from a study with this number of subjects and lack of details in medication usage records, we have discovered no “smoking gun” cause of VIIP in crew subjects. We have however, identified some medications used by crew that could be exacerbating factors in VIIP and have prepared recommendations for discussion with the VIIP Element and Space Medicine. INTRODUCTION: It has recently been reported that intracranial pressure (ICP) and/or vision changes, have occurred in a number of long duration astronauts. Some crewmembers have experienced changes in their vision after long-duration spaceflight on the ISS. These impairments include visual performance decrements, development of cotton-wool spots or choroidal folds, optic-disc edema, optic nerve sheath distention, and/or posterior globe flattening with varying degrees of severity and permanence. These changes are now used to define the visual impairment/intracranial pressure (VIIP) syndrome. The reasons for these potentially debilitating medical issues are currently unknown. The potential role of administered medications has not yet been examined, but it is known that many medications can have side effects that are similar to VIIP symptoms. Some medications raise blood pressure, which can affect intracranial pressure. Many medications that act in the central nervous system can affect intracranial pressures and/or vision. About 40% of the medications in the ISS kit are known to cause side effects involving changes in blood pressure, intracranial pressure and/or vision. For this reason, we proposed an investigation of the potential relationship between ISS medications and their risk of causing or exacerbating VIIP-like symptoms. The purpose of this study was to use medication usage records for affected and unaffected crew to determine if use of particular medications seemed to correlate with VIIP occurrence or severity. Due to the limited amount of data available from crewmembers, we added a large terrestrial data set to this study. Using publicly available FDA (Food & Drug Administration) Adverse Event Reports (FDA AERs) from many medications and medication classes, we identified increased reports of vision or intracranial pressure changes in several medication categories, which are presumed to be the most likely medications that could be involved in VIIP occurrence or severity. METHODS: This study employed a 2 -fold strategy to evaluate the role of medications in crewmembers' visual problems. First, the NASA LSDA (Life Sciences Data Archive) and LSAH data repositories were queried for crew health information regarding in-flight use of medications by all crewmembers who received sufficient testing to be assigned a CPG severity score. Use of medications was correlated in affected crew compared to the control group (CPG = 0) to assess the role of these medications in vision and ICP changes seen in spaceflight. Secondly, the FDA Adverse Event Reports were used a terrestrial source of data. These are publicly available and represent hundreds of thousands of cases in each quarterly data set. The quarterly reports were compiled into a database format and queried for data of interest. This resource represents a huge database; incorporating it into our study was an attempt to compensate for the extremely small sample number in the NASA data archive. For this data mining study, data associated with adverse events involving vision changes and intracranial pressure changes was examined. Data from NASA crew: This protocol was approved by the Johnson Space Center (JSC) Institutional Review Board (IRB), and all crewmember participants provided consent. LSAH provided medication usage data from a total of 37 individuals on 39 missions: 27 ISS crewmembers on 29 ISS missions and 12 Shuttle crewmembers who were on missions of less than 30 days. The crew medication use data available to us were frequently missing certain desired elements, most often, dates of use, or number of does taken. See Wotring, FASEB J 2015 for additional details on the medication use reports and their limitations. Every assumption we were forced to make was in the direction of an underestimation of medication usage, and most of the errors in the current data recording system are also in the direction of under-reporting.
Bibliography Type:	Description: (Last Updated: 12/24/2019)
Articles in Peer-reviewed Journals	Wotring VE. "Medication use by U.S. crewmembers on the International Space Station." FASEB J. 2015 Jul 17. [Epub ahead of print] Published online before print July 17, 2015. http://dx.doi.org/10.1096/fj.14-264838 ; PubMed PMID: 26187345 , Jul-2015